

UNIT – I

CONDUCTING MATERIALS

Classical free electron theory – Merits and Limitations - Expression for electrical conductivity - Fermi- Dirac distribution function – Density of energy states – Energy bands in Solids- Electron effective mass (Concept only) – Concept of hole – Computed Tomography Scan

INTRODUCTION

The physical properties of a metal are determined by its elementary charged particle called electron. The mobility of these electrons in the metals determines their electrical, thermal, and other properties. The electron theory of solids explains the structure and properties of solid on the basis of quantum theory. This is applicable to all type of solids whether metals or non-metals.

In this chapter we will restrict our discussion to the conducting properties of metal using the free electron model. The topics that will be discussed include, the density of energy states, Fermi-Dirac distribution function, effective mass of electron and drug delivery.

ELECTRON THEORY OF SOLIDS

Electron theory of solids explains the following concepts.

- Structural, electrical, magnetic and thermal properties of materials.
- This theory also explains the bonding in solids behaviour of conductors, semiconductors and insulators.
- The theory has been developed in three main stages.

CLASSICAL FREE ELECTRON THEORY

This theory, formulated by Drude and Lorentz around 1900, proposes that metals contain electrons capable of moving freely within the crystal lattice. It is a macroscopic model grounded in classical mechanics. In this framework, an electron's total energy corresponds entirely to its kinetic energy. The theory follows Maxwell-Boltzmann statistics, allowing all free electrons the possibility of acquiring energy.

QUANTUM FREE ELECTRON THEORY

In 1928, Sommerfeld introduced a microscopic theory that incorporated quantum mechanics and Fermi-Dirac statistics to describe the behavior of free electrons in metals. According to this quantum free electron theory, only a limited number of electrons can absorb energy, and it assumes that electrons travel under a uniform potential.

ZONE THEORY (OR) BAND THEORY

Also in 1928, Bloch developed the band theory, which describes electron motion within a periodic potential field created by the atomic lattice. This framework accounts for key concepts such as holes in the lattice, the formation of energy band gaps, and the effective mass of electrons—distinctive aspects of solid-state behavior. Additionally, band theory provides an explanation for the phenomenon of superconductivity.

BASIC TERMS INVOLVED IN THE FREE ELECTRON THEORY

Conductors

- The materials which conduct electricity when potential difference is applied are called electrical conductors.

Electric field

- It is the force experienced by a unit positive charge in the conductor.

$$E = \frac{F}{q} \quad \text{Unit: } NC^{-1} \qquad E = \frac{V}{L} \quad \text{Unit: } Vm^{-1}$$

Where V-potential difference

L- length of the material

Drift Velocity (V_d)

- The drift velocity is defined as the average velocity acquired by the free electron in particular direction, due to the applied electric field.

$$V_d = \frac{\lambda}{\tau}$$

$$V_d \propto E \Rightarrow V_d = \mu E \qquad \text{Unit: } ms^{-1}$$

Mobility (μ)

- The mobility is defined as the drift velocity (V_d) acquired by the electron per unit electric field (E).

$$V_d \propto E \Rightarrow V_d = \mu E \qquad \mu = \frac{V_d}{E} \qquad \text{Unit: } m^2V^{-1}s^{-1}$$

Mean free path (λ)

- The average distance travelled by an electron between two successive collisions in the presence of applied electric field is called mean free path.

$$\lambda = V_d \tau_c \quad \text{Unit: } m$$

Mean collision time (τ_c) (or) Collision time

- It is the time taken by the free electron between two successive collisions.

$$\tau_c = \frac{\lambda}{V_d} \quad \text{Unit: second}$$

Relaxation time (τ_r)

- It is the time taken by the electron to reach equilibrium position from disturbed position in the presence of electric field.

$$\tau_r = \frac{\lambda}{V_d} \quad \text{Unit: second}$$

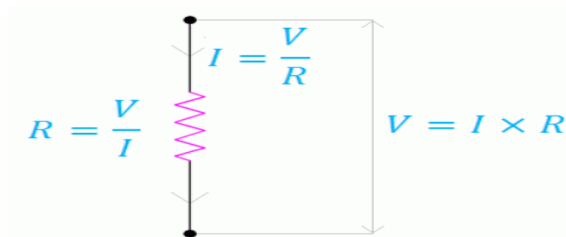
- Where λ is the distance travelled by the electron. The value of relaxation time is of the order of 10^{-14} sec. In the case of metals $\tau_r = \frac{\lambda}{V_d} = \tau_c$

Band gap (E_g)

- Band gap is the energy difference between the minimum energy of conduction band and the maximum energy of valence band.

Ohm's law

- The potential difference between any two points in a conductor is proportional to the current flowing through it.



$$V \propto I \Rightarrow V = IR$$

Resistance

- The opposition offered by the material to the motion of the electric charges is known as electrical resistance.

$$R \propto \frac{L}{A} \Rightarrow R = \frac{\rho L}{A} \quad \text{Unit: ohm } (\Omega)$$

Where ρ – Electrical resistivity

Current

The rate of flow of charges in a conductor is called current.

$$I = \frac{dq}{dt} \quad \text{Unit: A(or) } Cs^{-1}$$

Electrical resistivity

- Resistance offered by the material of unit area cross section and unit length is called as electrical resistivity.

$$\rho = \frac{\text{Resistance} \times \text{Area}}{\text{Length}} = \frac{RA}{L} \quad \text{Unit: ohm m } (\Omega m)$$

Factors affecting Electrical resistivity of metals

- Effect of temperature
- Effect of impurities
- Effect of stress
- Effect of plastic formation
- Effect of age formation

Current density (J)

- It is defined as the current per unit area of cross section of the current carrying conductor.

$$J = \frac{I}{A} \quad \text{Unit: } Am^{-2}$$

Conductance

- The reciprocal of electrical resistance is called as conductance.

$$G = \frac{1}{R} = \frac{A}{\rho L} \quad \text{Unit: } \Omega^{-1} (mho)$$

Electrical conductivity

- The reciprocal of electrical resistivity is called as electrical conductivity.

$$\sigma = \frac{1}{\rho} = \frac{L}{RA} \quad \text{Unit: } \Omega^{-1} m^{-1} (mho m^{-1})$$

CLASSICAL FREE ELECTRON THEORY

Assumptions:

- A Solid metal consists of positive ion cores; with the valence electrons move freely among these ion cores.
- The electrons move freely like molecules in a perfect gas.
- The total energy of the free electron inside a metal is equal to their kinetic energy.
- The total negative charge of electron is equal to the positive charge of ion cores and hence, the metal remains electrically neutral.
- This theory obeys Maxwell-Boltzman statistics which permits all free electrons to gain energy.
- The movement of free electrons obey classical kinetic energy theory of gases.
- In the absence of electric field ($E = 0$), the free electrons move in random directions and collide with each other. During this collision no loss of energy is observed since the collisions are elastic as shown in figure.
- In the presence of electric field ($E \neq 0$) the free electrons are accelerated in the direction opposite to the direction of applied electric field, as shown in figure.

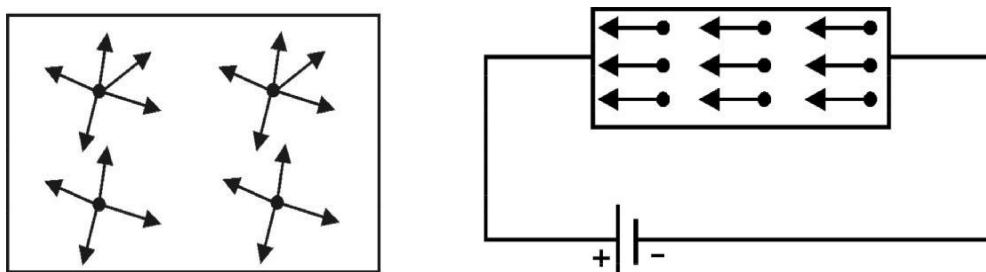


Fig2.1 Absence of electric field ($E=0$) Presence of electric field ($E \neq 0$)

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Mobility (μ)

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$$V_d \propto E \Rightarrow V_d = \mu E \quad \mu = \frac{V_d}{E} \quad \text{Unit: } m^2 V^{-1} s^{-1}$$

EXPRESSION FOR ELECTRICAL CONDUCTIVITY

Introduction:

Electrical conductivity is an important property of materials that describes their ability to conduct electric current. In conductors, the flow of electrons gives rise to electrical conduction. The electrical conductivity of a material depends on factors such as the number of charge carriers, their charge, and their mobility. The expression for electrical conductivity helps in understanding how current flows through a material when an electric field is applied. It plays a significant role in the study of electronic devices, semiconductor physics, and electrical engineering applications.

Definition

The amount of electric charges conducted per unit time across the unit area of cross section of the solid per unit applied electric field strength is called electrical conductivity.

$$\sigma = \frac{q}{tAE} = \frac{J}{E} \quad \left[J = \frac{q}{tA} = \frac{I}{A} \right] \quad \text{Unit: } \Omega^{-1} m^{-1} (mho \, m^{-1})$$

Derivation

Consider a metal rod of cross sectional area A and length L . Let ' n ' be the number of free electrons per unit volume and ' $-e$ ' be the charge of the electron and ' m ' be the mass of the electron.

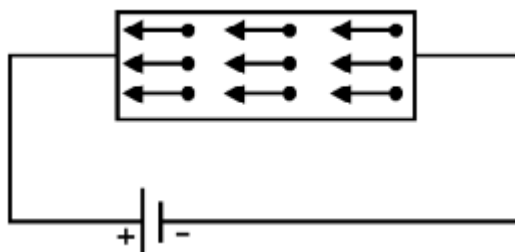


Fig. Electrical Conductivity

In the absence of electric field, free electrons move in random direction. When an electric field (E) is applied to the conductor the free electrons are accelerated and flows in a direction opposite to the applied electric field and give rise to current (I).

From Newton's second Law of motion, the force acquired by the electrons can be written as

$$F = ma \quad \dots\dots\dots(1)$$

When an electric field of strength (E) is applied to the conductor, the force experienced by the free electrons is given by

$$F = -eE \quad \dots\dots\dots(2)$$

Comparing equation (1) & (2)

$$ma = -eE$$

$$a = -\frac{-eE}{m} \quad \dots\dots\dots(3)$$

We know that, $\text{Acceleration} = \frac{\text{Drift velocity}}{\text{Relaxation time}}$

$$a = \frac{v_d}{\tau} \quad \dots\dots\dots(4)$$

From equations (3) & (4) we have

$$\frac{v_d}{\tau} = \frac{-eE}{m} \quad \dots\dots\dots(5)$$

$$v_d = \frac{-eE}{m} \tau \quad \dots\dots\dots(6)$$

We know that, Current density

$$J = -nev_d \dots\dots\dots(7)$$

Substituting equation (6) in equation (7)

$$J = -ne \left[\frac{-eE\tau}{m} \right]$$

$$J = \frac{ne^2\tau}{m} E \dots\dots\dots(8)$$

We know Current density

$$J \propto E$$

$$J = \sigma E \dots\dots\dots(9)$$

Compare equations (8) and (9)

$$\sigma = \frac{ne^2\tau}{m} \dots\dots\dots(10)$$

The equation (10) is known as electrical conductivity of a material. The electrical conductivity of a material is directly proportional to the free electron concentration in the material.

MERITS OF CLASSICAL FREE ELECTRON THEORY

- It is used to verify Ohm's law.
- It is used to explain electrical conductivity (σ) and thermal conductivity (K) of metals.
- It is used to derive Widemann-Franz law.
- It is used to explain the optical properties of metal.

LIMITATIONS OF CLASSICAL FREE ELECTRON THEORY

- It is a macroscopic theory.
- According to classical free electron theory, all the free electrons will absorb energy, but quantum free electron theory states that only few electrons will absorb energy.
- This theory cannot explain the Compton effect, Photo-electric effect, para-magnetism and ferromagnetism, etc.,
- This theory cannot explain the electrical conductivity of semiconductors and insulators.
- Dual nature of light radiation cannot be explained.

- The theoretical and experimental values of specific heat and electronic specific heat are not matched.
- The Lorentz number obtained by classical theory does not have good agreement with experimental value and theoretical value, but it was rectified by quantum theory.

FERMI – DIRAC DISTRIBUTION FUNCTION

Introduction

The Fermi–Dirac distribution function is a fundamental concept in quantum statistics that explains the distribution of electrons among available energy states. It obeys the Pauli Exclusion Principle. This distribution function is widely used to study the electrical and thermal behavior of electrons in metals and semiconductors.

Statement

The probability of finding an electron in a particular energy state at absolute temperature is given by

$$F(E) = \frac{1}{1 + \exp\left[\frac{E - E_F}{K_B T}\right]}$$

Where $F(E)$ – Fermi-Dirac Distribution Function

E – Energy of the energy level whose occupancy is to be determined

E_F – Fermi energy level

T – Absolute temperature

K_B – Boltzmann constant

The probability value $F(E)$ lies between 0 and 1.

- If $F(E) = 1$, the energy level is occupied by an electron.
- If $F(E) = 0$, the energy level is not occupied by an electron or vacant.
- If $F(E) = 0.5$ or $1/2$, then there is a 50% chance for the electron occupying in that energy level.

Effect of Temperature on Fermi Function

Case 1: At $T = 0K$ & $E < E_F$

$$F(E) = \frac{1}{1 + \exp\left[\frac{E - E_F}{K_B T}\right]}$$

$$F(E) = \frac{1}{1 + \exp\left[\frac{-ive}{0}\right]} = \frac{1}{1 + \exp(-\infty)} = \frac{1}{1 + 0}$$

$$F(E) = 1$$

It means that 100% probability for the electrons to occupy the energy level below the Fermi energy level.

Case 2: At $T = 0K$ & $E > E_F$

$$F(E) = \frac{1}{1 + \exp\left[\frac{E - E_F}{K_B T}\right]}$$

$$F(E) = \frac{1}{1 + \exp\left[\frac{+ive}{0}\right]} = \frac{1}{1 + \exp(+\infty)} = \frac{1}{1 + \infty} = \frac{1}{\infty}$$

$$F(E) = 0$$

It means that 0% probability for the electrons to occupy the energy level above the Fermi energy level.

Case 3: At $T = 0K$ & $E = E_F$

$$F(E) = \frac{1}{1 + \exp\left[\frac{E - E_F}{K_B T}\right]}$$

$$F(E) = \frac{1}{1 + \exp\left[\frac{0}{0}\right]} = 0.5$$

Case 4: At $T > 0K$ & $E = E_F$

$$F(E) = \frac{1}{1 + \exp\left[\frac{E - E_F}{K_B T}\right]}$$

$$F(E) = \frac{1}{1 + \exp\left[\frac{0}{K_B T}\right]} = \frac{1}{1 + \exp(0)} = \frac{1}{1 + 1} = \frac{1}{2}$$

$$F(E) = 0.5$$

It means that 50% probability for the electrons to occupy the Fermi energy level. At $T > 0K$ energy states above E_F are filled and below E_F are empty. Femi level is defined as the energy level at which the probability of occupation is $\frac{1}{2}$ at $T > 0K$.

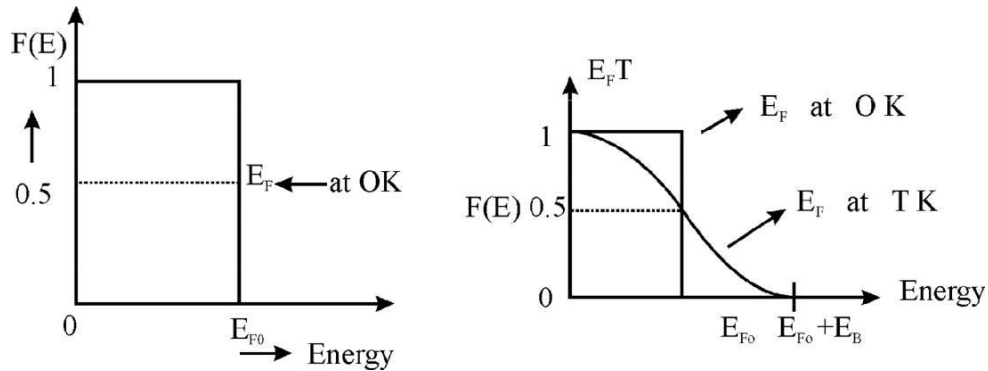


Fig. Fermi Dirac distribution function

Fermi level, Fermi Energy and their importance

- Fermi energy level is the maximum energy level of the quantum state up to which the electrons can be filled at absolute zero.
- It is the reference energy level which separates the filled energy levels and vacant energy levels.

Significance of $F(E)$

- To calculate the number of free electrons per unit volume at given temperature.
- To determine the Fermi energy of a metal.
- Fermi energy determines the energy of the particle at any temperature.

DENSITY OF ENERGY STATES

Introduction

The ability of a metal to conduct electricity depends on the available energy levels and the number of quantum states occupied by electrons. Therefore, it is necessary to study the energy states available for electrons, since they act as charge carriers in the material.

Definition

Density of states is defined as the number of energy states per unit volume in an energy interval E and $E+dE$ of a metal.

$$Z(E)dE = \frac{N(E)dE}{V}$$

Derivation

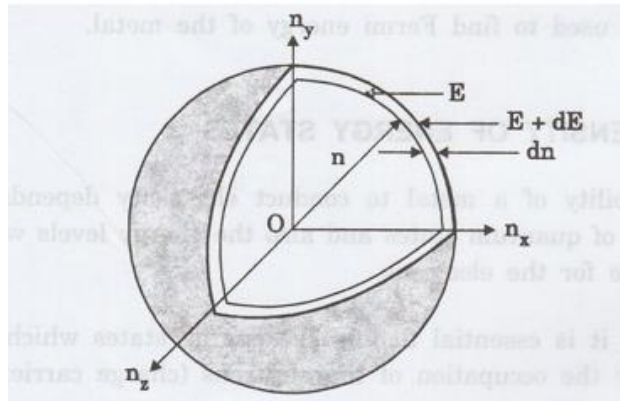
Consider a cubical metal specimen of side (L). Inside the cube, assume a sphere of radius (n) in space constructed with quantum numbers n_x, n_y and n_z .

$$n^2 = n_x^2 + n_y^2 + n_z^2$$

The sphere is further divided into many shells. Each shell represents a particular combination of quantum numbers and represents a particular energy value.

Therefore, the number of energy states within a sphere of radius (n) = $\frac{4}{3}\pi n^3$

Consider two shells of energy values E and $E + dE$. The number of energy states between the two shells can be calculated by finding the number of energy states between the shells of radius n and $n + dn$ from the origin. Since the quantum numbers are positive integers, n values can be defined only in the positive octant of the n-space.



The number of available energy states within the sphere of radius “n” due to one octant

$$= \frac{1}{8} \left[\frac{4}{3} \pi n^3 \right] \quad \text{-----(1)}$$

The number of available energy states within the sphere of radius “n + dn” due to one octant

$$= \frac{1}{8} \left[\frac{4}{3} \pi (n + dn)^3 \right] \quad \text{-----(2)}$$

The number of available energy states between the shells of radius n and $n + dn$ is

$$N(E)dE = \frac{1}{8} \left[\frac{4}{3} \pi (n + dn)^3 \right] - \frac{1}{8} \left[\frac{4}{3} \pi n^3 \right]$$

$$N(E)dE = \frac{1}{8} \times \frac{4}{3} \pi [(n + dn)^3 - n^3]$$

$$N(E)dE = \frac{1}{8} \times \frac{4}{3} \pi [n^3 + 3n^2 dn + 3n dn^2 + dn^3 - n^3]$$

Since the higher powers of dn is very small, dn^2 and dn^3 terms can be neglected.

$$N(E)dE = \frac{1}{8} \times \frac{4}{3} \pi [3n^2 dn]$$

$$N(E)dE = \frac{\pi}{2} [n^2 dn] = \frac{\pi}{2} [n(ndn)] \quad \text{-----(3)}$$

We know that the energy of the electron is given by

$$E = \frac{n^2 h^2}{8mL^2} \quad \text{-----(4)}$$

$$n^2 = \frac{8mL^2 E}{h^2} \quad \text{-----(5)}$$

$$n = \left[\frac{8mL^2 E}{h^2} \right]^{1/2} \quad \text{-----(6)}$$

Differentiating equation (5) we get

$$2n dn = \frac{8mL^2}{h^2} dE$$

$$n dn = \frac{8mL^2}{2h^2} dE \quad \text{-----(7)}$$

Substituting equations (6) and (7) in eqn(3) we get

$$N(E)dE = \frac{\pi}{2} \left[\frac{8mL^2 E}{h^2} \right]^{3/2} \frac{8mL^2 dE}{2h^2}$$

$$N(E)dE = \frac{\pi}{4} \left[\frac{8mL^2}{h^2} \right]^{1/2} E^{1/2} dE$$

According to Pauli's exclusion principle, each electron energy level can accommodate two electrons.

$$N(E)dE = 2 \times \frac{\pi}{4} \left[\frac{8mL^2}{h^2} \right]^{1/2} E^{1/2} dE$$

$$N(E)dE = \frac{\pi}{2} \left[\frac{8mL^2}{h^2} \right]^{3/2} E^{1/2} dE$$

$$N(E)dE = \frac{\pi}{2} \left[\frac{8m}{h^2} \right]^{3/2} L^3 E^{1/2} dE$$

Density of energy states $Z(E) dE = \frac{N(E)dE}{V}$

$$Z(E) dE = \frac{\frac{\pi}{2} \left[\frac{8m}{h^2} \right]^{3/2} L^3 E^{1/2} dE}{L^3} \quad \text{Since } V = L^3$$

$$Z(E)dE = \frac{\pi}{2} \left[\frac{8m}{h^2} \right]^{3/2} E^{1/2} dE$$

$$Z(E)dE = \frac{4\pi}{h^3} [2m]^{3/2} E^{1/2} dE$$

This equation represents density of energy states between the energy interval E and $E + dE$.

ENERGY BANDS IN SOLIDS

- Energy band theory of solids plays a very important role in determining whether a solid is a conductor, insulator or a semiconductor.
- In an isolated atom, electrons are very closely packed around the central positively charged nucleus and move along distinct orbits.
- The number of electron at the outermost orbit is called valence electrons.
- In the outermost orbits, the attractive force exerted by the nucleus on the electrons is relatively weak, making it easier for these electrons to be detached from the atom.
- The electrons that are easily detached from the outermost orbits are known as **free electrons**. In contrast, the electrons in the innermost orbits are strongly attracted to the positively charged nucleus and remain tightly bound to it; hence, they are referred to as **bound electrons**.
- A convenient method of representing the energy of different orbits is known as **energy levels**.
- The group of such large number of energy levels is called energy band.

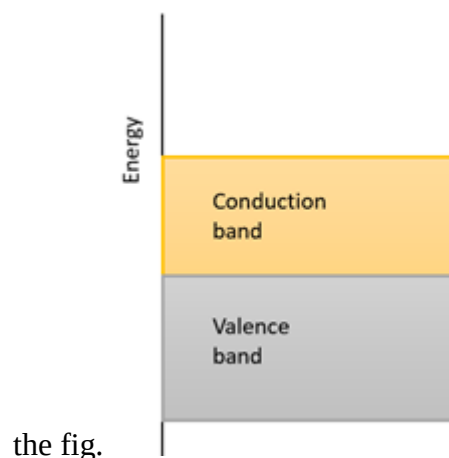
- Allowed energy levels in an energy band are almost continuous. These energy bands are separated by ranges of energies that have no allowed energy levels. These regions are known as forbidden band or energy gaps.
- The energy band made up of energy levels having valence electrons is called valence band. The second allowed energy band in a solid above the valence band is called conduction band. Electrons in this energy band are called conduction electrons. The conduction and valence bands are separated by a gap is known as forbidden energy gap.

Classification of Metals, Semiconductors and Insulators based on Band Theory

Based on the width of the forbidden energy gap between the valence band and the conduction band the solids are classified into insulators, semiconductors and conductors.

Conductors:

- A conductor is a material with high electrical conductivity that allows electric current to flow easily due to the presence of a large number of free electrons.
- There is no energy gap (E_g) between valence and conduction bands in case of conductors. The valence and conduction bands overlap with each other as shown in



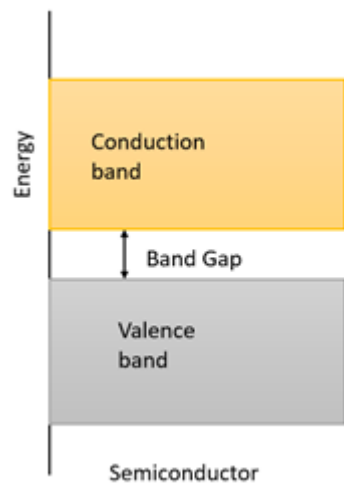
the fig.

- Electrons present in the valence band have the freedom to move into the conduction band, thus giving rise to the presence of numerous free electrons for conduction.
- The current (I) flowing through the conductor is made up of electrons. The resistivity of conductors is low and is of the order of $10^{-8} \Omega\text{m}$ to $10^{-6} \Omega\text{m}$.
- Example: Cu, Ag, Fe etc.

Semiconductors:

- Semiconductor is a material where electrical conductivity lies between insulators & conductors.

- The forbidden energy gap (E_g) in semiconductors is very narrow, as illustrated

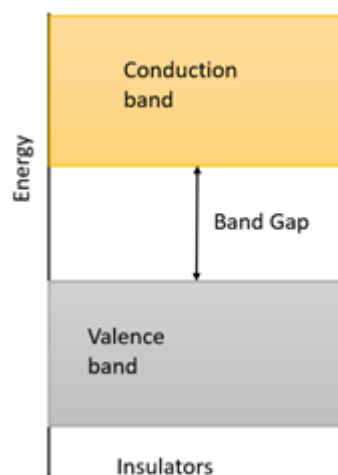


in the figure.

- At absolute zero temperature (0 K), the conduction band of a semiconductor is completely empty, while the valence band is fully occupied by electrons.
- As the forbidden gap is quite small, any small amount of energy from outside will be enough for electrons to get excited from valence band to conduction band.
- Resistivity of semiconductors is medium and is of the order of $10^{-5} \Omega\text{m}$ to $10^6 \Omega\text{m}$.
- Example: Ge($E_g=0.7\text{eV}$), Si($E_g=1.1\text{eV}$), etc.

Insulators

- An insulator is a material with very low electrical conductivity that does not allow electric current to flow easily because it has very few or no free electrons available for conduction.
- In insulators, the forbidden gap E_g is very large (greater than about 3 eV) as shown in fig.



- The electrons do not have sufficient energy to cross the large forbidden energy gap in insulators.
- Therefore, the electrons cannot be excited from the valence band to the conduction band, resulting in a lack of free charge carriers that can conduct electricity.
- Due to this, the electrical resistivity of insulators is very high, usually in the order of $10^{-7} \Omega\text{m}$ or more.
- Example: Diamond ($E_g = 6 \text{ eV}$) & Glass ($E_g = 10\text{eV}$)

EFFECTIVE MASS OF ELECTRON

When an electron in a periodic potential is accelerated relative to the lattice in an electric field or magnetic field, the mass of the electron is not a constant but it varies. This varying mass is called effective mass.

Definition:

Effective mass of electron is defined as the mass of electron which is subjected to periodic acceleration and is denoted as m_e^* . The effective mass of an electron is represented as

$$m_e^* = \left[\frac{\hbar^2}{\frac{d^2 E}{dk^2}} \right] \dots\dots\dots(9)$$

Where,

- m_e^* = effective mass of the electron
- $\hbar$ = reduced Planck's constant
- E = electron energy
- k = wave vector

CONCEPT OF HOLE (OR) NEGATIVE MASS OF ELECTRON

In a semiconductor material, once the electron acquires enough energy, it gets excited and moves from the valence band to the conduction band, creating a void in the valence band known as the hole. In spite of being non-existent in the form of a particle, the hole acts like a positively charged particle.

Definition:

Hole is defined as the void created in the valence band due to the absence of an electron and acts as a positively charged carrier of electricity.

Since a hole is non-existent in the form of a particle, it acts as a positively charged particle due to the absence of a negatively charged electron. Whenever a neighboring electron comes to occupy the hole's place, it creates a void in the place it originally occupied. In such a manner, the hole seems to move in the crystal lattice structure in the opposite direction to that of the electrons.

Under the influence of an electric field, holes drift towards the negative terminal and help in generating the electric current. In this way, electrical conduction in semiconductors takes place due to the drift of free electrons in the conduction band and holes in the valence band.

COMPUTED TOMOGRAPHY SCAN**Introduction**

Computed Tomography (CT) scanning is an advanced diagnostic imaging method that produces detailed cross-sectional images of the body's internal structures, enabling accurate visualization for medical evaluation. It combines X-ray technology with computer processing to create accurate images of internal organs, bones, tissues and blood vessels. CT scanning is widely used in hospitals for diagnosis, treatment planning and medical research.

Definition

A Computed Tomography (CT) scan is a diagnostic imaging method in which multiple X-ray images are taken from different angles around the body and processed by a computer to produce cross-sectional images called slices.

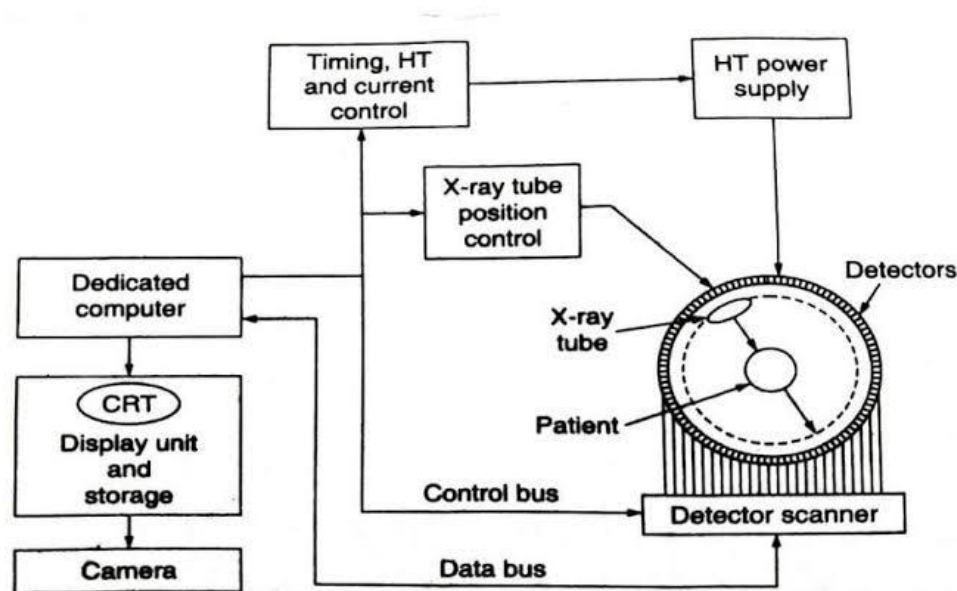
Principle of CT Scan

The fundamental principle of CT scanning relies on the varying degrees of X-ray absorption by different tissues in the body.

- X-rays penetrate the body.
- Different tissues absorb different amounts of radiation.
- Detectors measure the transmitted X-rays.
- A computer reconstructs the data into cross-sectional images.

Dense structures like bones absorb more X-rays and appear white, while soft tissues appear in shades of gray.

Block Diagram of CT Scan



Main Components

1. **X-ray Tube**
 - Produces X-rays that pass through the patient.
2. **Collimator**
 - Narrows and directs the X-ray beam.
3. **Patient Table**
 - Moves the patient through the scanner.
4. **Detectors**
 - Detect X-rays after passing through the body.
5. **Gantry**
 - Circular frame containing X-ray tube and detectors.
6. **Computer System**
 - Processes signals and reconstructs images.
7. **Display Unit**
 - Displays the CT images for diagnosis.

Working of CT Scan

1. The patient lies on the motorized table.
2. The table moves into the gantry.
3. The X-ray tube rotates around the patient.
4. X-rays pass through the body from multiple angles.
5. Detectors receive the transmitted X-rays.
6. Electrical signals are sent to the computer.
7. The computer reconstructs detailed cross-sectional images.
8. Images are displayed on the monitor for analysis.

Types of CT scans**1. Conventional CT**

- Takes one slice at a time.

2. Spiral (Helical) CT

- Continuous rotation of X-ray tube produces spiral images.

3. Multislice CT

- Produces multiple slices simultaneously for faster imaging.

4. Contrast CT

- Uses contrast dye for clearer visualization of blood vessels and organs.

Applications of CT scan**Medical Applications**

- Detection of tumors and cancers
- Brain imaging and stroke diagnosis
- Bone fracture detection
- Lung disease diagnosis
- Kidney stone detection
- Blood vessel imaging
- Guidance during surgery and biopsy

Engineering and Research Applications

- Biomedical research
- Image processing studies
- Radiation analysis

- Artificial intelligence in medical imaging

Advantages of CT scan

1. Produces highly detailed images
2. Fast and accurate diagnosis
3. Detects internal injuries quickly
4. Non-invasive procedure
5. Useful for soft tissue and bone imaging
6. Helps in treatment planning

Disadvantages of CT scan

1. Exposure to ionizing radiation
2. Expensive equipment
3. Risk of allergic reaction to contrast dye
4. Not recommended frequently for pregnant women
5. High maintenance cost